

How to turn your PS3 into a supercomputer

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Dartmouth

Gaurav Khanna, a University of Massachusetts Dartmouth physics professor, has created a step-by-step guide for building a homemade supercomputer — a feat he accomplished last year when he linked eight Sony PlayStation 3s together to compute complex calculations.

The gaming systems are connected by a central hub Khanna calls the “PS3 Gravity Grid,” which he said is 10-to-25 times more powerful than a typical computer. It helps with complicated tasks like one Khanna has been working on — finding the properties of gravitational waves generated by black holes.

The homemade computer makes is much cheaper than renting a supercomputer by the hour like many scientists do. A single simulation can take more than 5,000 hours at \$1 an hour, the university said.

“For the same cost, you can build your own supercomputer and it works just as well, if not better,” Khanna said. “Plus, you can use it over and over again, indefinitely.”

Khanna proposed his idea to Sony and last year received eight new PlayStations which run on an open platform, meaning they can be used like computers.

The how-to guide was created by Khanna and Chris Poulin, UMass Dartmouth’s principal investigator, who specializes in distributed pattern recognition and artificial intelligence.

The guide is available at www.ps3cluster.org. Khanna said the PS3 Gravity Grid can increase partnerships between UMass Dartmouth and other universities and businesses.

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